A. Data

Source Name:
NOAA Global Monitoring Laboratory (U.S. National Oceanic and Atmospheric Administration)

Full Source URL/s:
https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.txt
https://gml.noaa.gov/ccgg/trends/mlo.html

Annual mean atmospheric carbon dioxide (CO_2) concentration, measured in parts per million (ppm) at the Mauna Loa Observatory in Hawaii. This is the longest continuous direct record of atmospheric CO_2 (the Keeling Curve), starting in 1959.

Variables:

- Independent variable **x**: Year
- Dependent variable **y**: Annual mean CO_2 concentration (ppm)

There are well over 15 paired observations (which is roughly 1959–2025), and the relationship is approximately linear over this period (with a clear positive trend).

Year (x)	Annual Mean CO_2 Concentration (y, ppm)
1959.	315.98
1960.	316.91
1961.	317.64
1962.	318.45
1963.	318.99
1964.	319.62
1965.	320.04
1966.	321.37
1967.	322.18
1968.	323.05
1969.	324.62
1970.	325.68
1971.	326.32
1972.	327.46
1973.	329.68
1974.	330.19
1975.	331.13
1976.	332.03
1977.	333.84
1978.	335.41
1979.	336.84
1980.	338.76
1981.	340.12

1982.	341.48
1983.	343.15
1984.	344.87
1985.	346.35
1986.	347.61
1987.	349.31
1988.	351.69
1989.	353.20
1990.	354.45
1991.	355.70
1992.	356.54
1993.	357.21
1994.	358.96
1995.	360.97
1996.	362.74
1997.	363.88
1998.	366.84
1999.	368.54
2000.	369.71
2001.	371.32
2002.	373.45
2003.	375.98
2004.	377.70
2005.	379.98
2006.	382.09
2007.	384.02
2008.	385.83
2009.	387.64
2010.	390.10
2011.	391.85
2012.	394.06
2013.	396.74
2014.	398.81
2015.	401.01
2016.	404.41
2017.	406.76
2018.	408.72
2019.	411.65
2020.	414.21
2021.	416.41
2022.	418.53
2023.	421.08
2024.	424.61
2025.	427.35

B. Python Script

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import numpy as np
import matplotlib.pyplot as plt

# ============================================================
# 1. DATA
# Year (independent variable x) and annual mean CO2 concentration
# (dependent variable y) from NOAA Mauna Loa Observatory
# ============================================================
years = np.array([
1959, 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968,
1969, 1970, 1971, 1972, 1973, 1974, 1975, 1976, 1977, 1978,
1979, 1980, 1981, 1982, 1983, 1984, 1985, 1986, 1987, 1988,
1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998,
1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008,
2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018,
2019, 2020, 2021, 2022, 2023, 2024, 2025
])

co2 = np.array([
315.98, 316.91, 317.64, 318.45, 318.99, 319.62, 320.04, 321.37, 322.18, 323.05,
324.62, 325.68, 326.32, 327.46, 329.68, 330.19, 331.13, 332.03, 333.84, 335.41,
336.84, 338.76, 340.12, 341.48, 343.15, 344.87, 346.35, 347.61, 349.31, 351.69,
353.20, 354.45, 355.70, 356.54, 357.21, 358.96, 360.97, 362.74, 363.88, 366.84,
368.54, 369.71, 371.32, 373.45, 375.98, 377.70, 379.98, 382.09, 384.02, 385.83,
387.64, 390.10, 391.85, 394.06, 396.74, 398.81, 401.01, 404.41, 406.76, 408.72,
411.65, 414.21, 416.41, 418.53, 421.08, 424.61, 427.35
])

# Convert to float arrays for calculations
x = years.astype(float)   # independent variable
y = co2               # dependent variable
n = len(x)            # number of observations

# ============================================================
# 2. LEAST-SQUARES CALCULATIONS (manual formulas)
# ============================================================

# Compute the required sums
sum_x  = np.sum(x)      # Σx
sum_y  = np.sum(y)      # Σy
sum_xy = np.sum(x * y)   # Σxy
sum_x2 = np.sum(x**2)    # Σx²
```

```
# Slope a1 using the least-squares formula:
# a1 = (n*Σxy - Σx*Σy) / (n*Σx^2 - (Σx)^2)
a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x**2)

# Intercept a0 using the least-squares formula:
# a0 = (Σy - a1*Σx) / n
a0 = (sum_y - a1 * sum_x) / n

# ===============================================================
# 3. PREDICTED VALUES, RESIDUALS, AND GOODNESS-OF-FIT
# ===============================================================

# Predicted y values from the fitted line
y_hat = a0 + a1 * x

# Residuals = observed y - predicted y
residuals = y - y_hat

# Sr = SSE = sum of squared residuals (sum of squares of the errors)
Sr = np.sum(residuals**2)

# Total sum of squares (SST)
mean_y = np.mean(y)
SST = np.sum((y - mean_y)**2)

# Coefficient of determination r^2
r2 = 1 - (Sr / SST)

# Standard error of the estimate sy/x
syx = np.sqrt(Sr / (n - 2))

# Print the key results
print(f"Number of points n = {n}")
print(f"Intercept a0 = {a0:.6f}")
print(f"Slope     a1 = {a1:.6f}")
print(f"Sr (SSE)     = {Sr:.6f}")
print(f"r^2          = {r2:.6f}")
print(f"sy/x         = {syx:.6f}")

# ===============================================================
# 4. PREDICTION FOR A YEAR NOT IN THE DATA SET
# ===============================================================
x_pred = 2030.0
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```
y_pred = a0 + a1 * x_pred
print(f"Predicted CO2 in year {int(x_pred)} = {y_pred:.2f} ppm")

# ============================================================
# 5. GRAPHS
# ============================================================

# Graph 1: Data points + fitted straight line
plt.figure(figsize=(10, 6))
plt.scatter(x, y, color='blue', s=30, label='Observed data', zorder=3)
plt.plot(x, y_hat, color='red', linewidth=2,
    label=f'Fitted line: y = {a0:.2f} + {a1:.4f}x')
plt.xlabel('Year')
plt.ylabel('Annual mean CO2 concentration (ppm)')
plt.title('Linear Regression: Atmospheric CO2 at Mauna Loa (1959-2025)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('co2_fit.png', dpi=150)
plt.show()

# Graph 2: Residual plot
plt.figure(figsize=(10, 5))
plt.scatter(x, residuals, color='green', s=30)
plt.axhline(y=0, color='red', linestyle='--', linewidth=1.5)
plt.xlabel('Year')
plt.ylabel('Residual (ppm)')
plt.title('Residual Plot')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('co2_residuals.png', dpi=150)
plt.show()
```

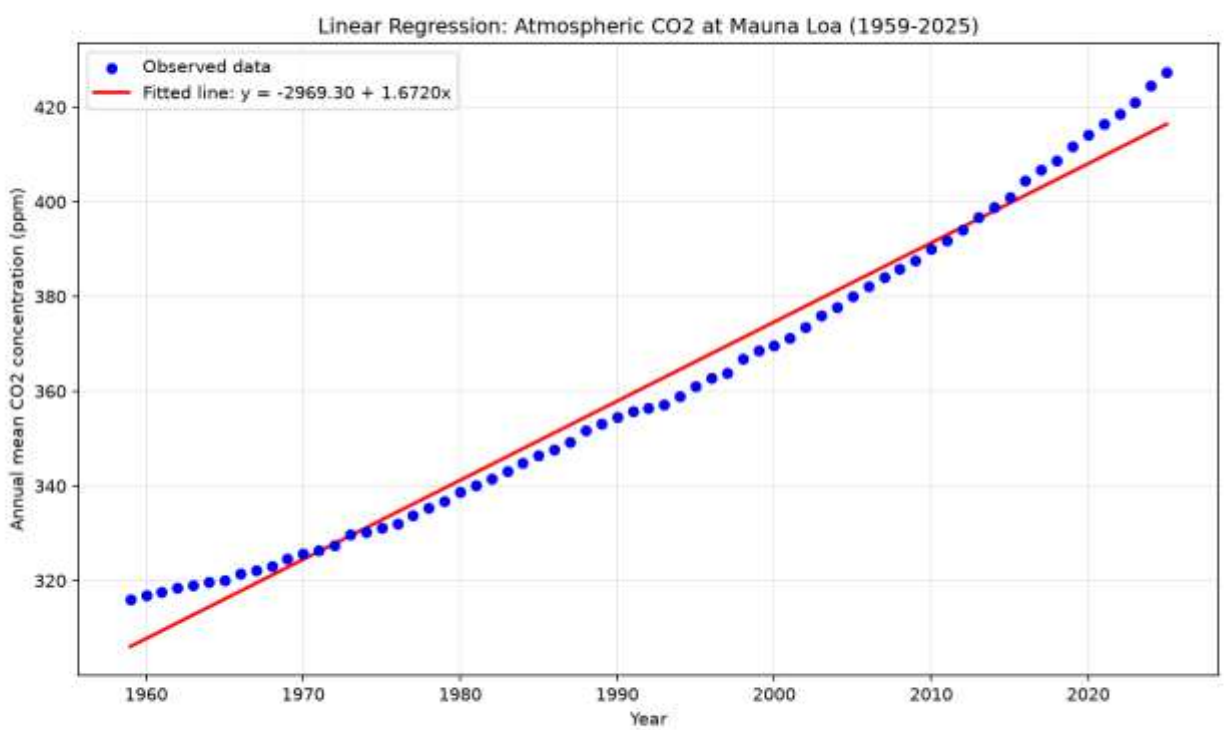
C. Regression Results

Results

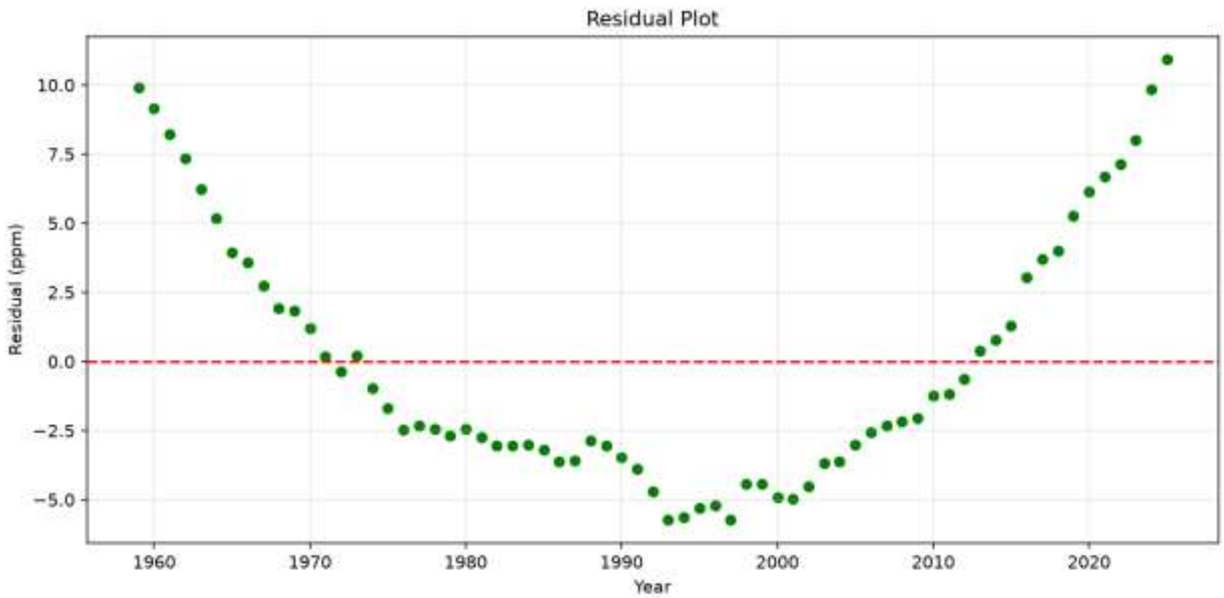
- **Regression equation**:
  $y = -2969.30 + 1.6720\, x$
  (where y is annual mean CO_2 in ppm and $x$is the year)
- **Intercept** $a_0 = -2969.30$ ppm
- **Slope** $a_1 = 1.6720$ ppm/year
- **Sr (SSE)** = 1389.08
- r^2= 0.9806
- **Standard error** $s_{y/x}$= 4.62 ppm

Graphs

- Data points with the fitted straight line:



- Residual plot:



Interpretation

- **Slope**: On average, atmospheric CO_2 concentration at Mauna Loa has increased by about **1.67 ppm per year** over the period 1959–2025.
- **Intercept**: The value –2969 ppm is the extrapolated CO_2 concentration at year 0. It has no physical meaning in this context; it is simply the mathematical intercept of the fitted line far outside the range of the data.
- **Goodness of fit**: $r^2 = 0.9806$means that the linear model explains about **98.1 %** of the variability in the annual CO_2 measurements. The standard error $s_{y/x} \approx 4.62$ppm indicates that the typical vertical distance of the data points from the fitted line is roughly 4.6 ppm. Overall, the straight-line fit is quite strong.
- **Residuals**: The residual plot shows a mild systematic pattern (residuals tend to be negative in the middle years and positive at both ends). This suggests that the true relationship is slightly curved (accelerating upward) rather than perfectly linear. Nevertheless, for a simple linear model the residuals remain relatively small compared with the overall rise of more than 110 ppm.

Prediction

Predict the annual mean CO_2 concentration for the year **2030** (a year not present in the data set):

$$\hat{y}(2030) = -2969.30 + 1.6720 \times 2030 \approx 424.79 \text{ ppm}$$

This prediction is meaningful because it gives a simple, data-driven estimate of the expected CO_2 level a few years into the future under the assumption that the long-term linear trend continues. (Note that because the real trend is accelerating, the linear model slightly under-predicts recent and near-future values)